

MPHY0047 Coursework 1 Report

Question 1 – Time-Based Statistical Analysis

Function

All statistics were calculated analytically using the standard definitions of mean, median, variance, skewness and kurtosis.

Mean:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad (1)$$

Variance

$$\sigma^2(x_i) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 \quad (2)$$

Skewness

$$\tilde{\mu}_3(x_i) = \frac{\sum_{i=1}^n (x_i - \bar{x})^3}{(n-1) \sigma^3} \quad (3)$$

Kurtosis

$$kurt(x_i) = \frac{\sum_{i=1}^n (x_i - \bar{x})^4}{(n-1) * \sigma^4} \quad (4)$$

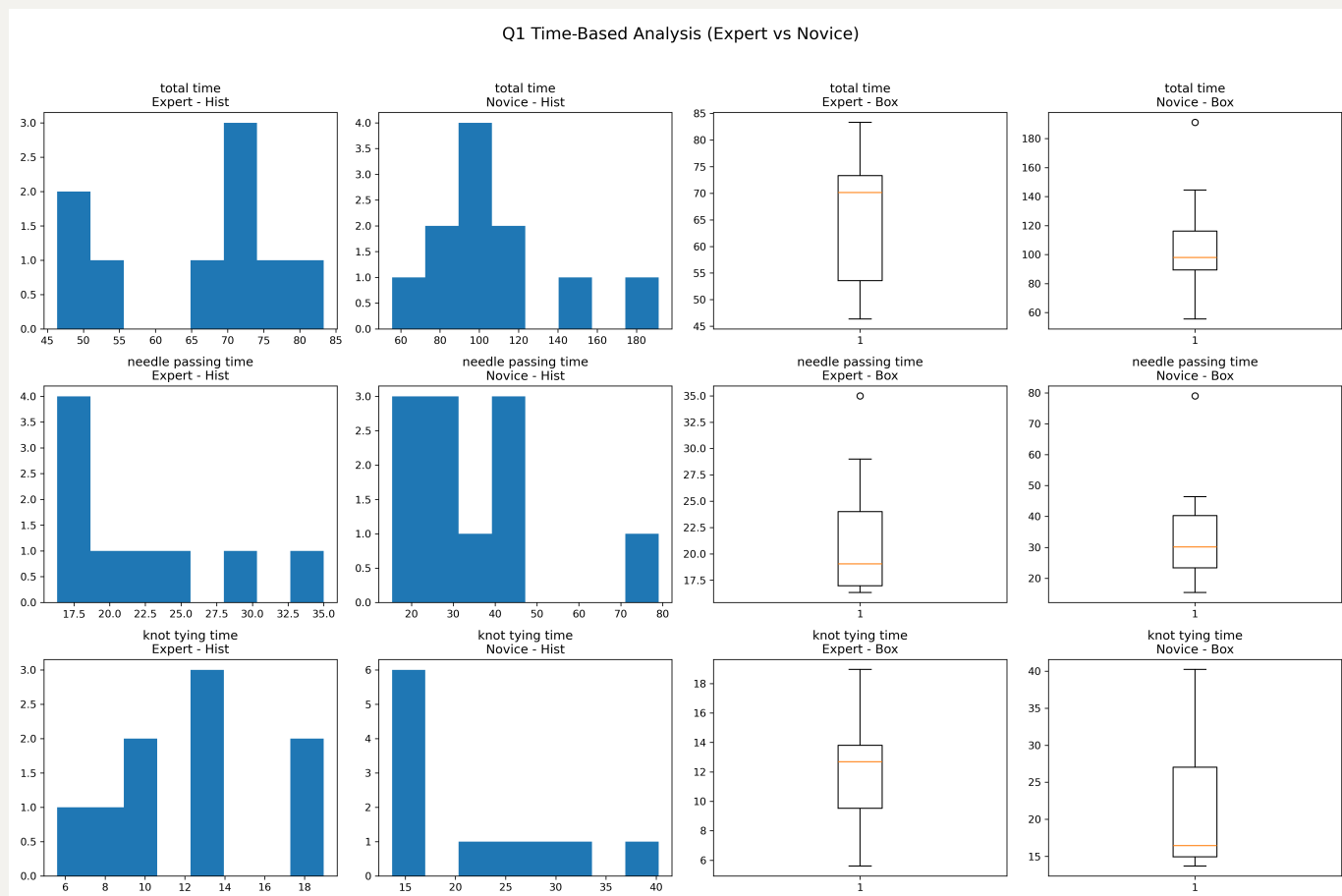
Statistical Summary

GROUP	METRIC	MEAN(S)	MEDIAN(S)	VARIANCE(s^2)	SKEWNESS	KURTOSIS
Expert	total time	65.116	70.12	158.474	-0.266	1.615
Novice	total time	107.073	98.08	1279.226	1.066	3.656
Expert	needle passing time	21.804	19.04	41.708	1.016	2.546
Novice	needle passing time	34.269	30.12	312.000	1.430	4.366
Expert	knot tying time	12.400	12.68	19.626	0.215	1.910
Novice	knot tying time	21.365	16.44	77.626	0.938	2.505

Across all three metrics, novices demonstrate longer completion times than experts. The difference is most pronounced for **total time** (107.073 s vs 65.116 s) and **needle passing time** (34.269 s vs 21.804 s), indicating reduced overall efficiency in the novice group.

Variance values are consistently higher for novices, particularly in total time (1279.226 vs 158.474), suggesting greater performance dispersion and lower consistency within this group. In addition, positive skewness in the novice data indicates the presence of several participants with notably prolonged completion times, especially for total time and knot tying time.

Among the three parameters, **knot tying time** shows the smallest absolute variance in both groups (19.626 for experts and 77.626 for novices), indicating comparatively lower dispersion. Therefore, it can be considered the most stable time metric across participants.



Analysis and Summary

From the boxplots, the novice group shows clear high-value outliers in total time and needle passing time, whereas the expert group does not present obviously. This suggests greater variability in the novice group.

Experts exhibit tighter interquartile ranges across all parameters, indicating more consistent performance.

Among the three parameters, knot tying time presents the smallest variance and the narrowest spread, suggesting higher robustness compared to total time. Therefore, knot tying time appears to be the most robust metric among the three time parameters.

Overall, total time shows the largest dispersion, needle passing time shows moderate dispersion, and knot tying time presents the smallest variability across participants.

Question 2

Statistical Summary Table

METRIC	U STATISTIC	P-VALUE	SIGNIFICANT($\alpha = 0.05$)	EFFECT (r)
total time	6.000	0.00109	Yes	0.879
needle passing time	22.000	0.04024	Yes	0.556
knot tying time	13.000	0.00622	Yes	0.737

Method Selection

Due to the small sample size (experts $n = 9$, novices $n = 11$), and the presence of skewed distributions and outliers observed in Question 1, the Mann–Whitney U-test was selected to compare differences between the two groups. The significance level was set at $\alpha = 0.05$.

Results Interpretation

According to the statistical test results:

Total time shows a statistically significant difference between the two groups ($p = 0.0011 < 0.05$). This indicates a clear difference in overall completion time between experts and novices. Combined with the mean results from Question 1, experts completed the task significantly faster, suggesting that total time has strong discriminative power.

Needle passing time also shows a statistically significant difference ($p = 0.0402 < 0.05$). This parameter can distinguish between the two groups, but the level of significance is weaker compared to total time, indicating relatively lower discriminative ability.

Knot tying time similarly demonstrates a statistically significant difference ($p = 0.0062 < 0.05$). This metric shows a clear distinction between groups, suggesting that this specific phase effectively reflects differences in experience level.

Conclusion

All three time parameters reached statistical significance, indicating that each can differentiate between experts and novices.

Combined with the findings from Question 1:

Total time shows the strongest discriminative ability.

Knot tying time demonstrates both statistical significance and relatively lower variability, indicating stable performance as a metric.

Needle passing time has comparatively weaker discriminative power.

Therefore, in the knot tying task, total time and knot tying time are the most representative discriminative indicators.

Question 3 – Error Metric Analysis

Statistical Summary

GROUP	N	MEAN ERROR	MEDIAN ERROR	VARIANCE
Expert	9	0.889	1.000	0.361
Novice	11	0.818	1.000	0.564

TEST	U STATISTIC	P-VALUE	SIGNIFICANT ($\alpha = 0.05$)	EFFECT (r)
Mann–Whitney U	53.000	0.79950	No	-0.071

Interpretation

The Mann–Whitney U test ($\alpha = 0.05$) indicates no statistically significant difference between experts and novices ($p = 0.7995$).

Therefore, the error metric does not significantly distinguish between the two groups in the knot tying task.

Conclusion

The error metric does not present statistical significance between the two groups. Therefore, unlike the time-based parameters in Question 2, the defined error metric does not effectively distinguish expertise level in the knot tying task.

Question 4 – Fixation Sparsity Analysis

Descriptive Statistics

Fixation sparsity was defined as the ratio of non-white pixels (fixation locations) to the total number of pixels (1920×1080 resolution).

The computed values are summarised in Table 4.

Table 4. Descriptive statistics of fixation sparsity

GROUP	N	MEAN	MEDIAN	VARIANCE
Expert	9	0.03437	0.03623	0.000066
Novice	11	0.05233	0.05164	0.000195

The novice group presents higher mean and median sparsity values, indicating a broader spatial distribution of gaze fixations.

Statistical Testing

Given the small sample size and potential non-normal distribution, the Mann–Whitney U test was applied.

Results:

- U statistic = 14.0
- p-value = 0.00784
- Effect size (rank-biserial r) = 0.717

Since $p < 0.05$, the difference between experts and novices is statistically significant.

Interpretation

The large effect size ($r = 0.717$) indicates a strong difference in fixation sparsity between the two groups.

This suggests that fixation sparsity is a meaningful metric for discriminating surgical expertise during knot tying.

Question 5 – Ranking of Discriminative Metrics

Based on the statistical outcomes from Questions 2–4, the five evaluated metrics are ranked according to their discriminative ability between expert and novice surgeons.

Ranking Table

RANK	METRIC	P-VALUE	EFFECT SIZE (r)	STATISTICAL SIGNIFICANCE
1	Total time	0.00109	0.879	Yes
2	Knot tying time	0.00622	0.737	Yes
3	Fixation sparsity	0.00784	0.717	Yes
4	Needle passing time	0.04024	0.556	Yes
5	Error metric	0.79950	-0.071	No

Justification

Total time presents the smallest p-value and the largest effect size, indicating the strongest discrimination between groups.

Knot tying time and fixation sparsity also show statistically significant differences with large effect sizes.

Needle passing time reaches statistical significance but demonstrates a comparatively smaller effect size.

The error metric does not reach statistical significance and therefore shows the weakest discriminative ability.

Conclusion

Among the evaluated parameters, total time appears to be the most effective metric for distinguishing expert and novice ophthalmic surgeons in the knot tying task.